

Specimen-current detector / 12 or 24 V input

Guarded LMC662 current acquisition / ADS1115

J1 SUM -> guarded LMC662 -> conditioning -> ADS1115.
100 Mohm feedback, 10 pF; all local voltages reference AGND.
Preserved 16 mm SUM trace; no copper plane under the input.

File: current12_analog.kicad_sch

9–30 V input / onboard buck / isolated converter

J2 regulated 9-30 V DC -> fuse -> reverse diode -> buck.
LMR51420XF: 500 kHz forced PWM; 5.08889 V nominal.
R05P09D/P provides isolated raw positive/negative rails.

File: current12_power.kicad_sch

Filtered local ± 5 V detector rails

Raw +/-9 V -> preload and RC filters -> TPS7A regulators.
Quiet +5VA and -5VA reference AGND; local 3.3 V for ADC.
55 mA per raw rail; verify ripple, headroom and temperature.

File: current12_power_rails.kicad_sch

POWER AND REFERENCE

Input: regulated 9–30 V DC, nominal 12 or 24 V.
Buck: 5.08889 V nominal; divider resistors 0.1%.
Analog: +5VA / –5VA and local 3.3 V.
AGND joins PGND only at R149 beside the converter input return.
ESP32 is powered separately. No gun or filament connection.

SIGNAL AND CONNECTORS

J1: 1 SUM specimen input, 2 AGND guard / shield.
J2: 1 VIN, 2 PGND. J4: both pins PGND / chamber.
J3: 1 GND, 2 reference output, 3 SDA, 4 SCL, 5 RDY.
SDA -> ESP32 GPIO21; SCL -> GPIO22; RDY optional.

MECHANICAL AND VALIDATION

170 x 100 mm, two copper layers, four 3.2 mm NPTH holes.
Use insulating nylon / PEEK fasteners and standoffs.
Native ERC/DRC and pin-map audits accompany the package.
These checks do not establish measured analog performance.

UNBUILT ENGINEERING PROTOTYPE

Validate leakage, noise, stability, settling and calibration.
Inspect construction and clean the high-impedance input.
No measured sensitivity or microscope resolution is claimed.
Commission first with a current-limited bench supply.
Check rails and detector baseline before microscope wiring.

Unbuilt engineering prototype / see ../README.md
Engineering prototype – see design notes and verified interfaces

Sheet: /
File: current12.kicad_sch

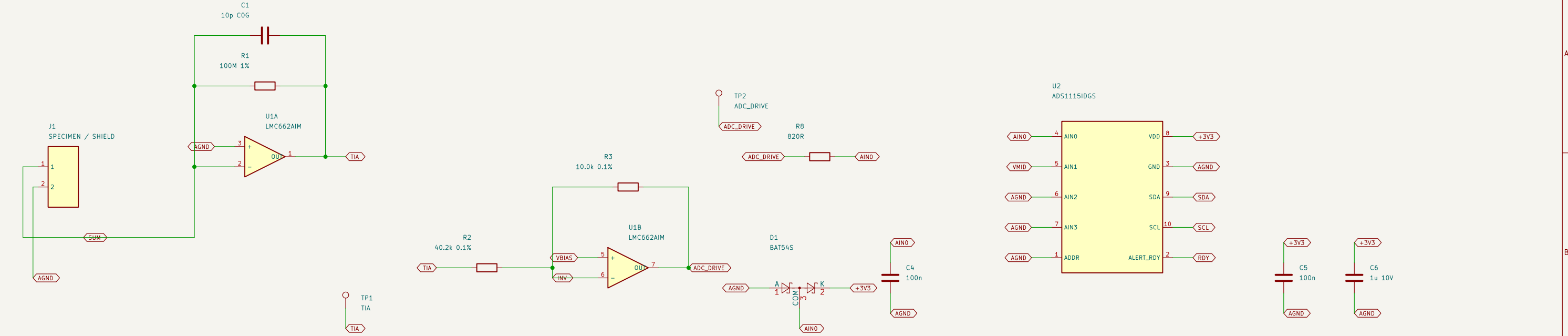
Title: Specimen-current detector / 12 or 24 V input

Size: A3	Date: 2026-09-13	Rev: B0 - UNBUILT
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01 GUARDED TRANSIMPEDANCE

02 ADC CONDITIONING

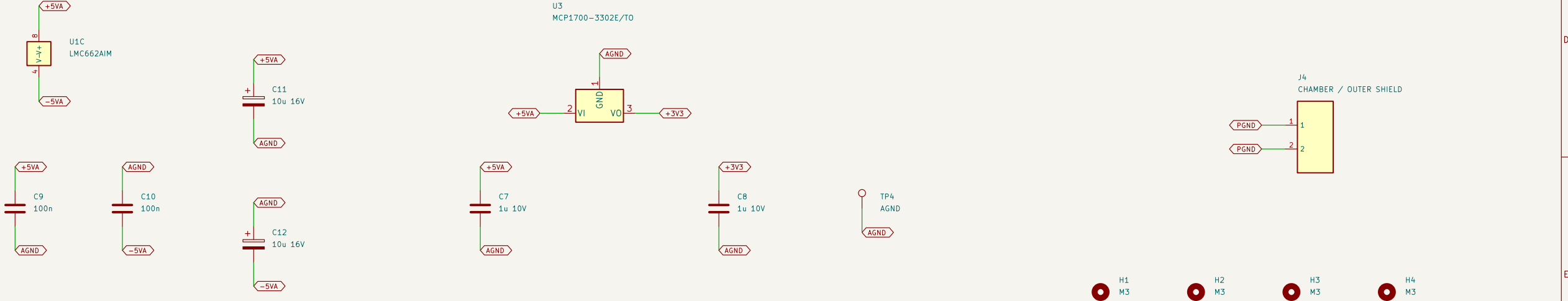
03 DIFFERENTIAL ADC



All voltages on this sheet reference AGND.
SUM has no clamp or probe point. R1 || C1: 1 ms nominal.
I_{abs} electron deposition gives V_{TIA} = +I_{abs} x 100 Mohm.
Do not connect SUM directly to the bias supply.

V_{A0} = 1.648358 - (10/40.2) V_{TIA}
V_{A1} = 1.650 V nominal. Dark-zero and gain calibration required.
PGA +/-256 mV gives 0.3141 pA per raw code; not measured sensitivity.

04 LOCAL BYPASS / QUIET ADC REFERENCE



UNBUILT PROTOTYPE
No measured leakage, noise, stability, settling, or imaging resolution.
LMC662 and feedback geometry preserved from the original detector.

Unbuilt engineering prototype / see ../README.md		
Engineering prototype - see design notes and verified interfaces		
Integrated detector prototypes		
Sheet: /Guarded LMC662 current acquisition / ADS1115/		
File: current12_analog.kicad_sch		
Title: Guarded LMC662 current acquisition / ADS1115		
Size: A3	Date: 2026-09-13	Rev: B0 - UNBUILT
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